	Glucose	Palmitate
Number of molecules oxidized	4	1
Total acetyl-coenzyme A generated	8	8
Total NADH generated	40	31
Total $FADH_2$ generated	8	15
Net ATP synthesized	128	106

Oxidation of a single palmitate molecule produces the same number of acetyl-coenzyme A molecules as oxidation of four glucose molecules. Based on the information in the table above, which statement best describes the use of oxygen when glucose and palmitate are oxidized for ATP production?

- A. Oxygen consumption per ATP synthesized is the same for oxidation of glucose and palmitate molecules. [23%]
- B. The number of oxygen molecules required for ATP synthesis is only dependent on the number of acetyl-coenzyme A molecules produced. [12%]
- C. Neither NADH nor $FADH_2$ is related to the number of oxygen molecules required for ATP synthesis. [19%]
- D. The number of oxygen atoms required per ATP synthesized is greater for palmitate oxidation than for glucose oxidation. [43%]

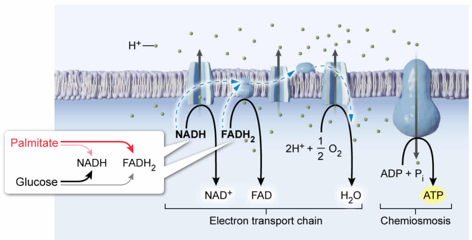
Correct
44% answered correctly

Explanation:

Oxygen consumption in ATP production

	Glucose	Palmitate
Number of molecules oxidized	4	1
Total acetyl-coenzyme A generated	8	8
Total NADH generated	40	31
Total $FADH_2$ generated	8	15
Net ATP synthesized	128	106

Total number of cofactors = total number of oxygen atoms consumed



Palmitate produces **more** $FADH_2$ & **less** NADH than glucose. $FADH_2$ electrons are used less efficiently, so more oxygen is required to produce ATP.

The energy required for mitochondrial ATP synthesis is drawn from the potential energy in the proton gradient across the mitochondrial inner membrane. This gradient is generated by the **electron transport chain** (ETC), which pumps matrix protons to the intermembrane space during electron transport and, in its final step, **uses matrix protons** and **O_2 for** the **synthesis of water**.

As shown in the table, the initial four glucose molecules produce the same amount of acetyl-coenzyme A (CoA) as palmitate. Each acetyl-CoA molecule produces the same amount of NADH and $FADH_2$ in the citric acid cycle. However, during the respective steps that convert glucose and palmitate to acetyl-CoA in the first place, metabolism of glucose produces NADH and ATP, whereas fatty acid metabolism produces NADH and $FADH_2$. Because NADH enters the ETC at complex I and $FADH_2$ enters the chain at complex II, fewer protons are pumped across the inner mitochondrial membrane in response to electrons from $FADH_2$ than from electrons donated by NADH.

The question asks for the statement that best describes the use of oxygen during ATP synthesis from glucose or palmitate oxidation. The data in the table indicate that oxidation of the four glucose molecules generates 48 cofactors, yielding a net production of 128 ATP. In contrast, palmitate oxidation generates 46 cofactors (almost the same number as glucose) and yields a net production of 106 ATP (significantly less than glucose). Because electron transfer from **each cofactor** results in **reduction of one oxygen** atom from O_2 but $FADH_2$ is less efficient, **oxygen consumption** is **greater during palmitate** oxidation **than** during **glucose oxidation**.

(Choice A) Oxygen consumption per ATP synthesized is greater for oxidation of palmitate than for oxidation of glucose.

(Choices B and C) The number of oxygen molecules required for ATP synthesis is determined by the number and type of reduced cofactors (ie, NADH and $FADH_2$) that donate electrons to the ETC during ATP synthesis.

Educational objective:
For each pair of electrons donated to the electron transport chain by NADH or $FADH_2$, one oxygen atom is consumed during the reduction of O_2 to H_2O. NADH yields more efficient ATP synthesis than $FADH_2$.

Time spent:0 sec
Subject:
Biochemistry

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System:
Metabolic Reactions